

Good evening everyone



class start at 8:10

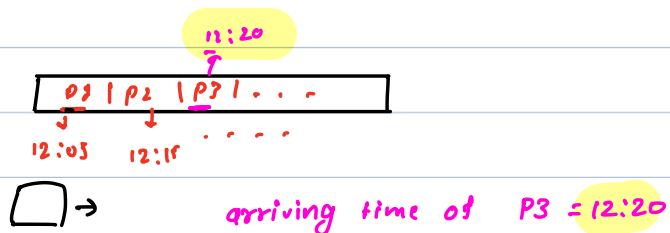
CPU scheduling Algorithm

it is responsible for determining the order in which process are executed.

P1 P2 P3 P4

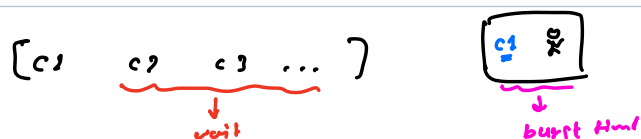
1) Arriving time (AT) :-

the arrival time of a process is the time at which it enters a ready queue.



2) Burst time (BT) :-

the amount of CPU time required by the process to finish its execution.



P3 takes 15 min.

3) **completion time (CT)**: it is the time at which the process finish its execution and exits the CPU.

P3 leaves the queue at 12:55

exit time

4) **turn around time (TAT)**

total time spend by the process.

it is difference between completion time and arrival time.

$$TAT = CT - AT$$

$$12:55 - 12:20 = 35 \text{ min}$$

5) **waiting time (WT)**: it is the difference between TAT and burst time.

$$\text{waiting time} = TAT - bt$$

$$35 \text{ min} - 15 \text{ min}$$

$$\text{waiting time} = 20 \text{ min}$$

FCFS (First come first serve)

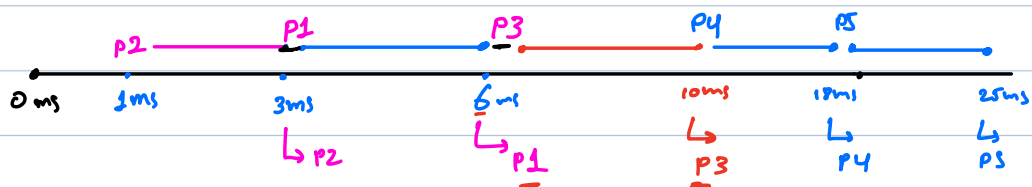
this is the simplest CPU algorithm
process are executed in the order they
arrive in ready queue.

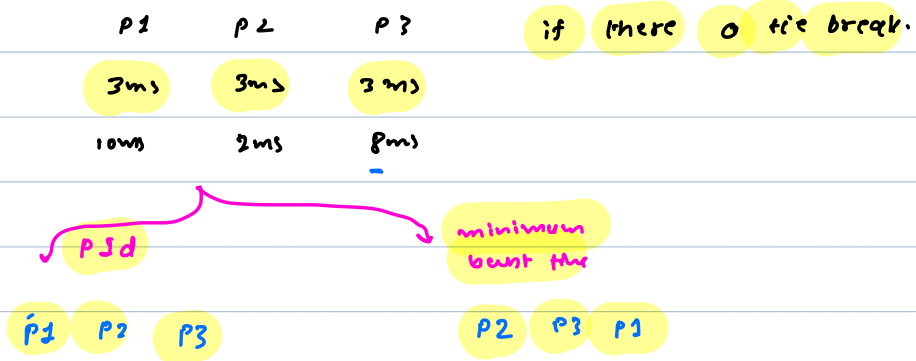
it is non-preemptive in nature
once a process execution, it continue until
the completion.

P1 | P2 | P3 ...

PID	Arriving time	burst time	Completion time	TAT ct-at	wt ct-bt
P1	2	3	6ms	4ms	1ms
P2	1	2	3ms	2ms	0ms
P3	3	4	10ms	7	3ms
P4	4	8	18ms	14	6ms
P5	5	7	25ms	20	13ms

P2 P3 P3 P4 P5





priority → FCFS

urgent process may have to wait.

P1	P2	P3
↓		
2000 min	5min	15min

convoy effects

it long process ahead of short ones.
causing them to wait excessively.

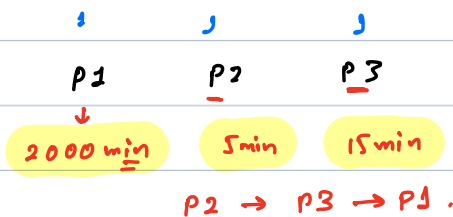


Shorter job first (SJF) / Shorter job next (SJN)

- it is non preemptive in nature.
- the process with shortest CPU burst time is selected for execution

shortest remaining time first (SRTF)

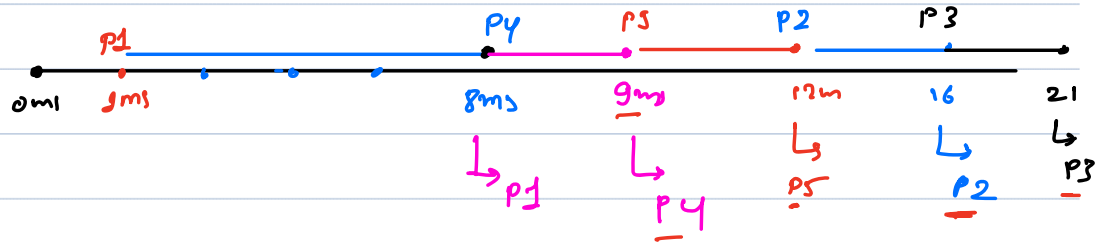
- it is preemptive in nature.
- if a new process arrives with a shorter time than the currently execution process, switch to new process.



PID	Arriving time	burst time	Completion time	TAT ct-at	wt ct-ft
P1	1	7	8		
P2	3	4	16		
P3	4	5	21		
P4	7	1	9		
P5	8	3	12		

SJF

~~P1~~ | ~~P2~~ | P3 | ~~P4~~ | P5



break

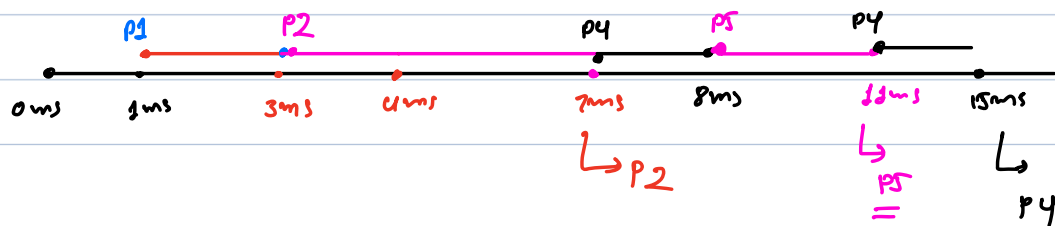
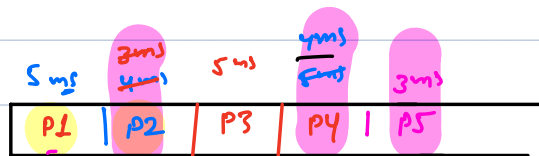
thu

9:18

PID	Arriving time	burst time	Completion time	TAT ct-at	wt ct-bt
P1	1	7	20ms		
P2	3	4	7ms		
P3	4	5	28ms		
P4	7	5	15ms		
P5	8	3	11ms		

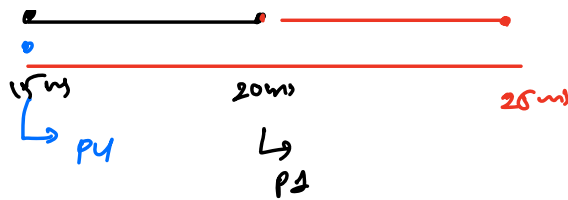
SRIF

preemptive

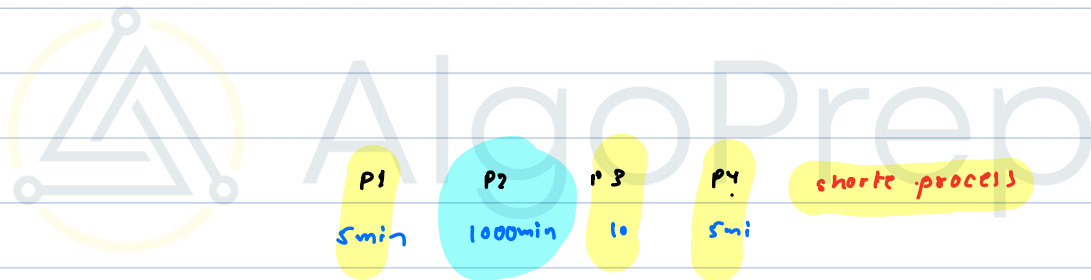


P1

P2



- minimum average waiting time and response time.
- require knowledge of CPU burst time in advance. which may not always be available.
- historical data
- predict burst time



starvation of long process

it refers to a situation in which a process unable to progress or complete.
it happen when continuously shorter process are arrive.

Round robin (RR)

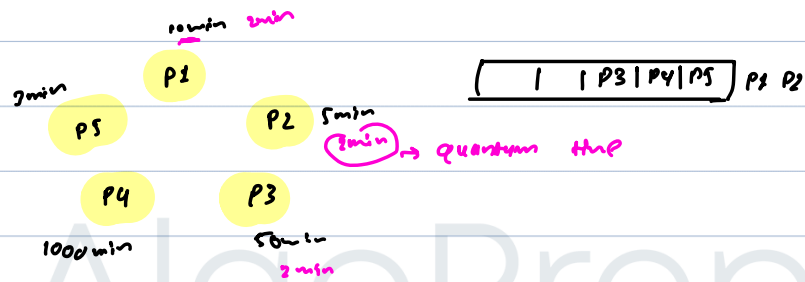
- it work in cyclic manner.

each process receives a small unit of time.

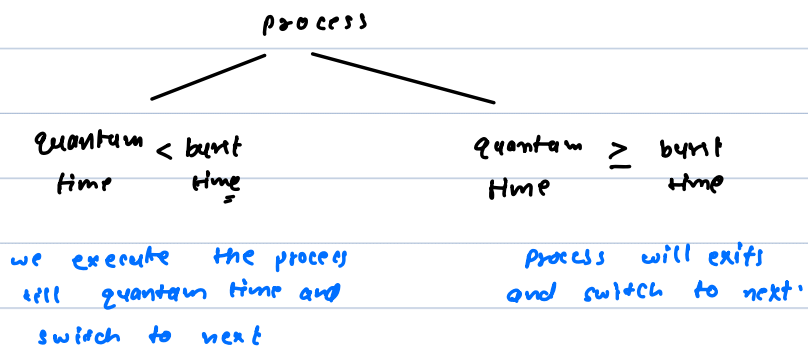
- it is called quantum time

- it is preemptive in nature.

there is no prioritised based selection



- each process executes for a limited time before being switch it placed back in ready queue.



PID	Arriving time	burst time	Completion time	TM9 ci-at	wt tAF-bl
P1	1	9			
P2	3	3			
P3	5	6			
P4	6	10			
P5	7	7			



Quantum time = 4



- if quantum time is minimum.
- context switching increase
- it leads to high overhead.

if quantum time is too large.

limited context switching.

it is great like FCFS